

Benchmark

MA.912.GR.2.5 Given a geometric figure and a sequence of transformations, draw the transformed figure on a coordinate plane.

Benchmark Clarifications:

Clarification 1: Transformations include translations, dilations, rotations and reflections described using words or using coordinates.

Clarification 2: Instruction includes two or more transformations.

Connecting Benchmarks/Horizontal Alignment

- MA.912.GR.3.2, MA.912.GR.3.3
- MA.912.GR.4.3
- MA.912.GR.6.5

Terms from the K-12 Glossary

- Coordinate Plane
- Dilation
- Origin
- Reflection
- Rigid Transformation
- Rotation
- Scale Factor
- Translation

Vertical Alignment

Previous Benchmarks

- MA.8.GR.2.1, MA.8.GR.2.2, MA.8.GR.2.3
- MA.912.F.2.1

Next Benchmarks

- MA.912.NSO.4
- MA.912.F.2.2, MA.912.F.2.3, MA.912.F.2.5
- MA.912.T.4

Purpose and Instructional Strategies

In grade 8, students learned about the effects of translations, rotations, reflections, and dilations on geometric figures. In Algebra 1, students extended this knowledge of transformations to transforming functions using tables, graphs and equations. In Geometry, students are given the preimage and a sequence of two or more transformations (a composition) and draw the image on a coordinate plane. In later courses, types of transformations will be expanded to include stretches that can transform functions and conic sections and conversions between rectangular and polar coordinates.

- The purpose of this benchmark is that students produce an image on the coordinate plane as the result of combining transformations (translations, dilations, rotations and reflections).
- Instruction includes transformations described using words and using coordinates.
- Instruction includes understanding that a single transformation can be done as a sequence of transformation. Likewise, some sequences can be described as a single transformation.
 - For example, if the transformation sequence is to translate 1 unit to the left, then rotate 90-degrees clockwise about the origin, then translate 1 unit to the right; the result is the same as rotating the figure 90-degrees clockwise about the point (1,0). This can be described in coordinates as $(x, y) \rightarrow (x - 1, y)$, then $(x -$

$1, y) \rightarrow (y, -(x - 1))$, then $(y, -(x - 1)) \rightarrow (y + 1, -(x - 1))$, and the result of rotating the figure 90-degrees clockwise about the point $(1,0)$ can be described in coordinates as $(x, y) \rightarrow (y + 1, -x + 1)$.

- Instruction includes discussing the resulting images using the same preimage and a set of transformations in different orders and understanding that transformations may or not be commutative based on the sequence and type of transformations.
 - For example, if the transformation is the result of a vertical and a horizontal translation, the sequence is commutative.
 - For example, if the transformation is the result of a 180° rotation and a reflection over the x -axis, the sequence is commutative.
 - For example, if the transformation is the result of a 90° rotation and a reflection over the x -axis, the sequence is not commutative.
- Instruction includes the understanding that if a dilation is part of a sequence, it can be applied at any time within the sequence of transformations without changing the final result.

Common Misconceptions or Errors

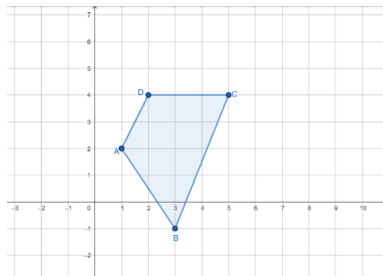
- Students may think order of transformations doesn't matter.

Instructional Tasks

Instructional Task 1 (MTR.3.1, MTR.5.1)

Part A. On the coordinate plane, draw the resulting figure after transforming quadrilateral $ABCD$ through the following sequence below.

- Reflect quadrilateral $ABCD$ over the line $y = x$.
- Translate horizontally and vertically the resulting figure using $(x, y) \rightarrow (x + 3, y - 2)$.



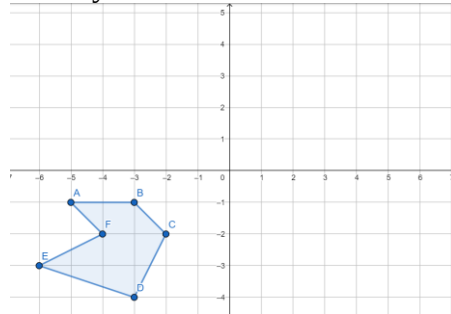
Part B. Would the resulting figure be the same if the transformations were reversed? How did you come to your conclusion?

Instructional Items

Instructional Item 1

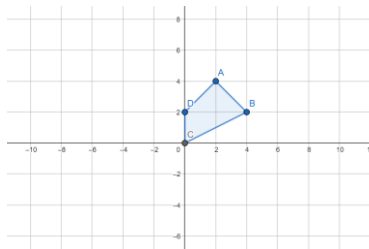
Perform the following sequence of transformations on the polygon $ABCDEF$ on the coordinate plane.

- Rotate 180° counterclockwise about the origin.
- Then, translate horizontally 2 units to the left and vertically 3 units down.



Instructional Item 2

Draw the resulting figure after quadrilateral $ABCD$ is transformed using $(x, y) \rightarrow (-x, y - 3)$.



**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*